Sparse One-Way LoRa Clock Transfer: Bandwidth Experiments, Authenticated Two-Step Timing, and a Fixed-500-kHz Validation

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Abstract

LoRa can extend a short-range timing deployment with a passive, sparse-update clock path, but packet detection delay, channel changes and oscillator prediction must be distinguished from useful clock phase. This retrospective reports T096/SX1262 experiments that progressed from narrow-band frequency hopping, through two- and five-bandwidth screens, to a fixed 922.4 MHz, 500 kHz, SF11, CR4/8 profile. An authenticated two-step exchange separates a captured transmit launch marker from its later timestamp, while a causal receiver model bridges nominal 30-second updates. A 500 kHz arm showed lower centered endpoint scatter in a small hopped screen; intermediate-band results were sparse and nonmonotonic. They motivated a fixed-profile validation rather than establish an optimal bandwidth. A fixed prior of +102.931911 ms, measured from four earlier pairs, was applied to the receiver's native model input before validation. All 900 declared future GNSS PPS opportunities were captured, eligible and mapped. Delivered-model phase had 269.140 ns RMS, −95.018 ns mean, 580 ns absolute P99 and 624 ns maximum. Six physical-SYNC-age bins each contained 150 observations and remained below 296 ns RMS. These are finite, GNSS-input-referenced phase results with a bench-specific prior, not independent electrical-output or portable UTC qualification. We preserve incomplete attempts, packet denominators and the limitations of one-way timing.

1 Motivation and relation to prior work

A local clock is useful between messages, not just at a successful packet. Sparse LoRa timing therefore joins two problems: estimating a well-defined radio event and predicting the receiver clock while the radio is silent. PToB normally distributes authenticated BLE time. Its LoRa extension seeks a receive-only fallback for places where BLE is unavailable, with no client request, acknowledgment or ranging transmission. This paper asks what the existing experiments establish about bandwidth, event delay and 30-second clock delivery. It reports their results and the selected wire format; it adds no device capture or experiment.

LongShoT demonstrated low-microsecond LoRaWAN synchronization using hardware/MAC timestamping and drift compensation [1]. It establishes relevant precedent for moving timestamps below application scheduling. Rinaldi et al. analyze preamble/SFD estimation from sampled chirps and the influence of sampling and frequency offsets [2]. Their simulated correlation architecture is different from a commodity radio's HeaderValid interrupt. Müller et al. use the SX1280's two-way ranging engine at 2.4 GHz, with oscillator and PLL hardware and external PPS comparisons [3]. Their time-deviation results and propagation compensation answer a different question from one-way SX1262 model phase. Neither LoRa modulation nor a small internal fit residual identifies a calibrated UTC error by itself.

The contribution here is an evidence-linked progression from raw endpoint characterization to a complete future-witness population, together with an explicit authenticated timing contract. Comparisons retain bandwidth, spreading factor, channel policy and measured quantity. Earlier unavailable and incomplete populations remain separate from the successful finite run.

2 Two-step timing and the selected wire format

2.1 Physical markers and causal clock delivery

The measured boards are Heltec T096/HT-n5262G devices with an nRF52840 MCU, SX1262 radio and UC6580 GNSS receiver. A 16 MHz capture clock has 62.5 ns ticks. For transmission, a hardware event route captures the first BUSY falling edge after the SetTx command commits. This is a launch-side marker, not an externally calibrated antenna RF onset. The receiver captures HeaderValid and preserves that event until the complete packet passes CRC and authentication. HeaderValid is later than first RF arrival; its latency depends on the modem configuration.

SYNC identifies a precision event; FOLLOW authenticates the original SYNC cursor and the launch marker mapped through its frozen source clock. Neither USB polling nor FOLLOW arrival defines the SYNC timestamp. Source withdrawal, capture-epoch changes and mismatched cursor/context invalidate associations. A source or emitter change requires fresh authenticated context and a fresh fit. The emitter may export qualified own-GNSS time or admitted locked BLE time; its physical binding is independent of BLE grandmaster election. Bindings and keys do not select a runtime mesh.

The receiver fits only LoRa observations. Up to 16 pairs, at least 20 s apart, are retained over 300 s; at least four points spanning 180 s are required for training. The baseline affine rate fit may be replaced by guarded agreement of 180/300 s pair-slope medians. Phase is a causal median of historical projections at the selected rate. After FOLLOW, the owner commits at a current capture; a one-second phase slew preserves continuity for discrepancies up to 1 ms. Larger discontinuities reacquire. These internal guards are not uncertainty bounds. The GNSS witness used below does not enter this live estimator.

2.2 Wire version 3, schedule 8

All integers are little endian. Every packet begins with `4c 03`, a kind/flags byte and the schedule byte `08`. Kind is the low nibble; bits 4–5 are zero. HAS_PAIR (`80`) and TIME_VALID (`40`) are control-only flags. A pair hint requires valid time and a nonzero delay in 10 ms units. Schedule 8 sends no DISC packet, although kind 1 remains defined in the broader wire family. Incompatible schedules fail closed.

Packet	Bytes	Body after the four-byte header, followed by MAC16
META (2)	112	Emitter/source context, ordering and rendezvous metadata
SYNC (3)	24	Original physical cursor:u32
FOLLOW (4)	32	Original SYNC cursor:u32, SYNC launch UTC ns:u64

META offsets	Fields
4,12; 20,28; 36,44	Emitter/boot; root/boot; parent/boot, six u64 fields
52,56,60	Source epoch, emitter stream, public seed, three u32 fields
64,68,72	Cursor:u32, serial:u32, preparation UTC ns:u64
80,82,84,86	TZ:u16, offset minutes:i16, hint:u16, source kind:u8
87–95; 96–111	Reserved zero; authentication tag

META authenticates its 96-byte body with a group HMAC-SHA256 truncated to 128 bits, using domain `PTL3MET1`. SYNC and FOLLOW use `PTL3SYN1`/`PTL3FOL1` over the cached 16-byte context plus their complete body. Context domain `PTL3CTX1` binds a normalized META header, bytes 4–63, source kind and three zero bytes. Thus epoch, stream, emitter, root, parent and schedule remain inseparable from precision packets. META's preparation UTC is an ordering snapshot, not an RF-start timestamp. Source kind 1 denotes own GNSS and 2 selected BLE; own-GNSS context has zero parent fields. Symmetric membership authenticates a group member, not individual hardware. Original capture/expiry and replay floors constrain reuse; unknown independent source age prevents an absolute freshness guarantee.

3 Experimental progression: hopping and bandwidth

3.1 Why characterize the endpoint before the clock?

The initial narrow precision PHY used SF7, 41.67 kHz and CR4/6, with separate 915 MHz SF12/BW500 control rendezvous. A public seeded permutation selected 64 precision channels on a 902.4 MHz + $0.4i$ MHz raster. Cursor and authenticated context coordinated SYNC/FOLLOW tuning. Frequency hopping can diversify a channel-dependent path, but also mixes channel delays into a sparse clock fit. No controlled experiment here isolates an accuracy or reliability benefit of hopping, and no range result is inferred from narrow bandwidth.

A historical 1200.127 s hopped slow-profile capture retained 942 diagnostics: 476 delivered-available and 466 unavailable. It observed 23 new pairs; 8 of 22 pair intervals exceeded the 32.51 s normal-cadence bound. No local PPS witness was present. Nine causal 30 s future-pair comparisons gave 4.496 μs RMS for an old raw-anchor projection and 5.878 μs for the median/discipline projection, with three unknown and two censored endpoints. These noisy pair-consistency metrics neither demonstrated improvement nor measured independent delivered-clock phase.

3.2 The two-bandwidth screen

Run B283 alternated low/high arms in an ABBA ordering on the hopped raster, using SF7, CR4/6, preamble 12, explicit header and CRC for both. Each raw pair was bracketed by adjacent eligible local GNSS PPS. No phase correction was applied. All 15 offered SYNCs were CRC-valid and formed accepted bracketed pairs. The complete screen had 484 eligible PPS and no sequence gap, but only 219 PPS carried a delivered mixed-arm map; 265 were NO_MAP.

B283 arm	Offered/pairs	Mean phase (ms)	Raw RMS (ms)	Centered SD (ns)
41.67 kHz	8/8	−76.974486	76.974486	3743.567
500 kHz	7/7	−6.417616	6.417616	217.550

The high-bandwidth arm's lower centered scatter was a useful measured lead. Its raw phase still had millisecond bias. Different hop channels and sequential sampling confound bandwidth with channel/source variation; ABBA reduces a simple drift confound but cannot remove it. This small screen does not rank all bandwidths or establish sub-microsecond uncorrected clock accuracy.

3.3 Five bandwidths on common precision channels

B285 used SF7/CR4/6 and a forward/reverse five-band sequence at nominal 30 s cadence. Every bandwidth shared SYNC channel 16 (908.8 MHz) and FOLLOW channel 48 (921.6 MHz), reducing the earlier frequency confound. Two recovery streams remain separate. Their primary mean phases and centered population SDs are:

	Stream A			Stream B		
BW (kHz)	Offered/pairs	Mean (ms)	SD (ns)	Offered/pairs	Mean (ms)	SD (ns)
41.67	3/3	−76.972408	2532.629	1/1	−76.973311	—
62.5	2/2	−51.320425	889.053	2/2	−51.315249	57.336
125	1/1	−25.663754	—	2/2	−25.660492	625.150
250	2/1	−12.834315	—	2/2	−12.833635	923.304
500	1/1	−6.417797	—	2/1	−6.417361	—

A dash means one pair, insufficient to estimate scatter. All 18 SYNCs were CRC-valid; 16 produced bracketed pairs. Of 604 eligible PPS, 81 had a mixed-arm map and 523 did not; no sequence gap was observed. The small, nonmonotonic within-stream scatter prevents a general noise-versus-bandwidth law.

4 From endpoint bias to a held-out fixed-profile test

4.1 Selection of a fixed 500 kHz candidate

The raw mean phase in B285 changes strongly with inverse bandwidth. LoRa symbol duration is $T_s = 2^{\mathrm{SF}}/\mathrm{BW}$ [2]; packet recognition therefore introduces a configuration-dependent event separation. This motivates characterizing each endpoint configuration instead of treating HeaderValid as an unbiased arrival timestamp. It does not identify how much of the observed bias belongs to radio detection, launch timing or propagation. For example, B283's 217.550 ns centered SD and its 6.418 ms raw RMS describe different aspects of the same high-bandwidth arm.

The selected candidate fixed all packet classes at 922.4 MHz, 500 kHz, SF11, CR4/8, preamble 12, explicit header and CRC. Continuous reception and the absence of retuning remove hopping rendezvous and inter-channel delay as variables in this candidate. SF11/CR4/8 also changes symbol duration and coding from the SF7/CR4/6 screens. Consequently the final validation is a new system configuration, not a controlled test of bandwidth alone. Selection reflects the observed high-bandwidth scatter, a simpler fixed-channel association, and a feasible sparse airtime budget. Range, interference resistance and a globally best PHY remain unmeasured.

SYNC/FOLLOW are 24/32 bytes every 30 s; META is 112 bytes every 300 s; DISC is absent. Calculated packet airtimes are 263168, 295936 and 787456 μs, respectively. Adding 10 ms per packet gives a nominal guarded hour of

$$120(263168 + 295936 + 20000) + 12(787456 + 10000) = 79061952\ \mu\mathrm{s},$$

or 2.196165% of an hour. All recovery reservations also consume the 90 s (2.5%) per-boot rolling-hour cap. This calculation is not measured RF occupancy, a reboot-persistent limit or regulatory certification. Configured 18 dBm chip drive and nominal 28 dBm board output are settings/nominal values; physical power, antenna gain and EIRP were not measured.

The spreading-factor choice also follows the budget. With the same payloads, coding, preamble, header, CRC and cadence, SF12 would require 460800, 591872 and 1443840 μs per SYNC, FOLLOW and META. Its guarded 146.166720 s/hour (4.060187%) exceeds the cap. SF11 is therefore the highest feasible uniform spreading factor among these two candidates, with 10.938048 s/hour nominal headroom for recovery; this is a design calculation, not a measured range gain.

4.2 A prior calibration separated from validation

A raw fixed500 development capture, on earlier image .286, supplied the first four newly observed authenticated physical pairs with adjacent eligible PPS brackets. Their phases in ns were −102932551.855, −102931705.402, −102932025.397 and −102931797.225. The negative median, rounded to integer ns, was frozen as $d_0 = +102931911$ ns before the later receiver images and validation. Its raw four-pair centered SD was 328.443 ns. Four correlated pairs do not establish a calibration uncertainty or transferable delay.

For each accepted later FOLLOW, the actual estimator input is

$$u_k^{\mathrm{model}} = u_k^{\mathrm{transmitted}} + d_0.$$

The correction is applied before native fitting and delivered-map publication; authentication and freshness continue to use the original packet. The witness never supplies live phase/rate updates, filter tuning or per-window recentering. This is a measured bench phase base, incorporating the calibration endpoints and path. It is neither a time-of-flight measurement nor a distance correction. It is expected to require revalidation across hardware, PHY, temperature and placement changes.

An earlier .287 attempted validation remains incomplete: of 900 planned future PPS, 215 were captured, 206 had maps and nine were NO_MAP; 685 planned opportunities were uncaptured. The 150.578 ns RMS of the mapped subset is retained as conditional evidence, not the result of a complete 900-PPS trial and not pooled with .288. Earlier empty screens retained 244, 55, 44 and 30 eligible PPS without a bracketed pair or delivered map; they supply no bandwidth precision estimate.

5 Complete fixed-profile validation

5.1 Population and measured quantity

The final experiment used matched T096 .288 sender/receiver images, the frozen prior and a BLE-disabled LoRa receiver. The receiver's local GNSS PPS was an independent input witness: it was not a live input to the LoRa estimator. After the first observed trained map, a declared sequence boundary selected the next 900 PPS opportunities, sequences 28–927. It was based on model readiness, not future residuals. The complete capture contained 927 eligible and mapped PPS; the primary population is only the declared future 900. All 900 were captured, reference-eligible and mapped, with zero PPS sequence gaps or NO_MAP outcomes. Sender-side transmission-offer coverage was unknown; receiver completeness must not be rephrased as perfect packet delivery.

At each retained local PPS input edge, the firmware evaluated the currently delivered LoRa map, including its actual phase slew. The phase statistic is modulo a GNSS PPS second. Native whole-second GNSS association remained unknown. A separate observed nearest-second/RMC agreement did not calibrate absolute UTC. Thus the witness is separate from estimator training, while capture/map evaluation and a previously calibrated endpoint still form part of the measurement chain. No independent emitted PPS or downstream load was measured.

Population	n	Mean (ns)	RMS (ns)	Centered SD (ns)	Abs. P99 (ns)	Max. (ns)
Declared future .288	900	−95.018	269.140	251.809	580	624
Incomplete .287, mapped	206	72.816	150.578	131.801	326	354

Every future .288 phase is retained without residual rejection, post-hoc offset, detrending or smoothing. P99 is the nearest-rank percentile of absolute phase; SD uses the population denominator. All 900 sampled absolute phases were below 1 μs. This finite observation does not bound the continuous trajectory between PPS edges or predict rare tails in a longer deployment.

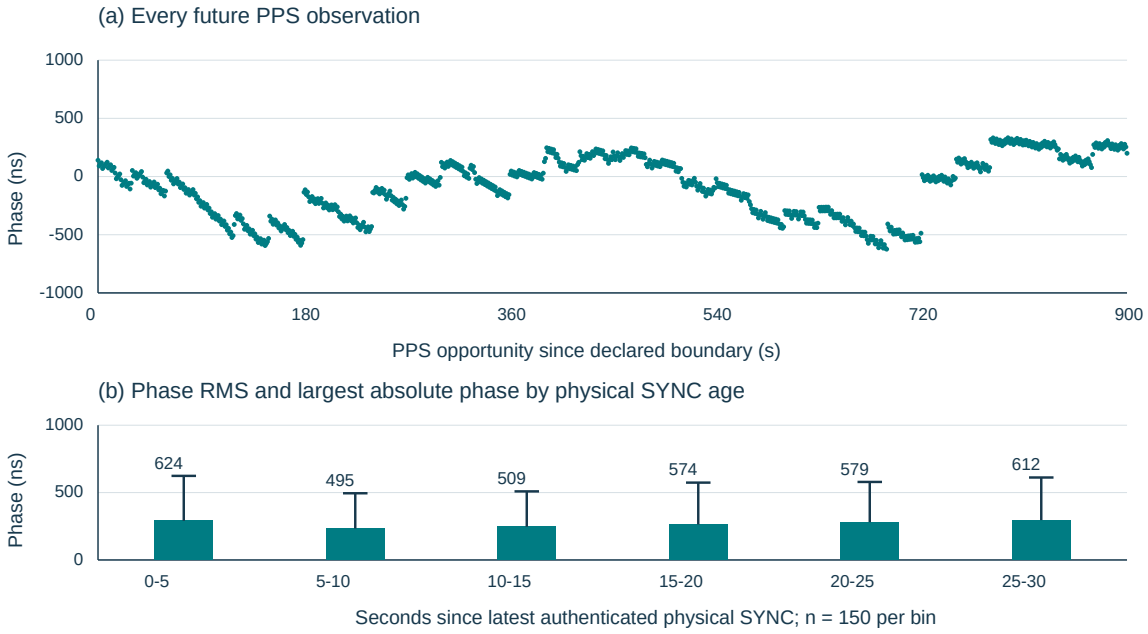


Figure 1: Complete declared .288 validation. (a) All 900 delivered-model phases at GNSS PPS inputs; the vertical scale includes ±1 μs. (b) RMS bars and maximum-absolute caps in six physical-SYNC-age bins, each with 150 observations. No tail or interval is removed.

The phase scatter is visibly correlated across updates, consistent with a shared causal prediction rather than 900 independent replications. The full 269.140 ns RMS, rather than the 251.809 ns centered SD, is the principal phase summary after the independently frozen prior.

6 What the interval results establish

6.1 Physical SYNC age and usable-map age

Two clocks of age matter. Physical-SYNC age measures time since the most recent authenticated physical SYNC, even while its FOLLOW is still in flight. Accepted-pair age measures time since the SYNC belonging to the map's last committed pair. Resetting the first age does not magically update the delivered clock. During FOLLOW latency, the receiver still extrapolates the previous map. Keeping these coordinates separate avoids making early-in-interval performance look artificially fresh.

Physical SYNC age (s)	0–5	5–10	10–15	15–20	20–25	25–30
PPS observations	150	150	150	150	150	150
Phase RMS (ns)	290.580	232.961	246.404	263.105	280.258	295.679
Maximum absolute (ns)	624	495	509	574	579	612

All six RMS values remain below 296 ns. There is no missing physical-age bin; none of the primary observations had unknown age or physical-SYNC age at least 30 s. Separately, 91 samples had accepted-pair age at least 30 s, because a new SYNC preceded its completed FOLLOW. Those samples had 311.897 ns RMS and 617 ns maximum. Their inclusion demonstrates that the low aggregate RMS does not simply discard the previous-map portion of a precision interval.

This is stronger evidence for the tested sparse clock than a low fit residual at packet arrival. It is still sampled, calibrated evidence from one future window. No outage or extended holdover performance follows from six well-filled bins under continuing successful updates. Source permission and the original physical/context expiration fences remain necessary even when extrapolation looks smooth.

6.2 Bandwidth choice and the scope of "best observed"

Within B283's matched SF7/CR4/6 screen, 500 kHz had the lower observed centered endpoint scatter. In B285, the pooled descriptive SDs across two recovery streams were 2227.904, 2663.568, 1619.841, 819.229 and 218.137 ns from low to high bandwidth. The increase from 41.67 to 62.5 kHz and the tiny per-stream populations prevent a monotonic law. In particular, each stream's 500 kHz arm had only one accepted pair, and the pooled two-pair SD is not a stable noise estimate. Pooling cannot restore a missing complete forward/reverse block.

The final .288 run is the strongest complete, independently PPS-witnessed delivered-model population selected here: 269.140 ns RMS over all 900 declared future opportunities. Calling it the best complete observed result is a statement about this evidence population. Its RMS cannot be ranked as the same statistic as raw pair centered SD, in-sample link-fit RMS, future-pair consistency or time deviation in other work. The incomplete .287 subset even has a smaller conditional RMS; its missing opportunities prevent substituting it as the primary result.

6.3 Availability beyond the final T096 window

The .288 window does not erase context turnover or heterogeneous service gaps. A separate 1800-snapshot extension on a different receiver retained all snapshots but reported LoRa availability in only 267; its 265–1512 ns internal fit consistency had no independent phase witness. Such observations motivate availability and handover qualification, not a transfer of the T096 result to another target. Similarly, later source/receiver images share wire fields without inheriting the measured .288 accuracy.

7 Limits, reproducibility and next measurements

The central identifiability limit is one-way transfer: a clock offset, a fixed endpoint delay and propagation appear together. A constant prior can make local model phase agree with a prior bench reference while leaving portable UTC accuracy unresolved. A complete error budget would need independent source uncertainty, GNSS input/capture delay, TX-marker-to-RF delay, RX-detection delay, antenna/cable paths and the actual output path. The present data do not identify each term. A low timer tick, MAC validity or a checksum is insufficient evidence of timing accuracy.

The result also leaves environmental and service questions open. There is one complete final window, one earlier four-pair calibration, no controlled bandwidth-by-distance campaign, no field coverage distribution, no measured EIRP and no interference/thermal population. Repeated held-out windows across placements and temperatures would test calibration stability. A common clock and external time-interval instrument at transmitter/receiver outputs would separate input-model agreement from independently measured output error. A prespecified channel/PHY matrix with sufficient repetitions would be needed to compare fixed and hopped operation while retaining failed rendezvous and all packet offers. These are proposed measurements, not work performed here.

The implementation boundary also matters. The fixed-profile cap is RAM-only per boot; unknown source age remains unknown; LoRa observations do not create BLE grandmaster, relay or USB timing authority. A longer observation lease is not a promise of delivery or a calibrated uncertainty. Continuous receive can simplify timing association, but its energy consumption has not been measured in this experiment. These qualifications leave the finite result useful: a sparse, authenticated one-way clock can retain sub-microsecond sampled phase through its 30-second cycle in this calibrated T096 configuration.

7.1 Reproducible selected evidence

The accompanying `papers/lora-sparse-timing/` directory contains canonical LaTeX/PDF, 900 .288 and 215 .287 opportunity rows, all 31 selected bandwidth pairs with per-context offer counts, and the four calibration phases. The standard-library analyzer recomputes means, uncentered RMS, centered SD, nearest-rank absolute P99, maxima, both age populations, prior median and packet airtime accounting. The vector plot draws every .288 observation. Provenance records the selected private-input hashes and redactions; no raw console log, key, enrollment, image or physical identity is redistributed. The public derivatives reproduce the reported statistics, not private raw capture extraction or physical calibration. Source snapshots distinguish the measured .288 image from the later compatible wire reference; edition hashes establish document identity, not accuracy.

Author-review status and licensing. This is a draft for Jonathan Yantis to review before any public publication. The manuscript, original figures and original selected datasets are CC BY 4.0, credited to Jonathan Yantis; original analysis code is MIT. AI assistance supported evidence organization, analysis-code preparation and drafting; the author retains responsibility for interpretation and approval. No new hardware operations or experiments were performed to prepare this retrospective.

References

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